

Spending Maintenance App

Collaborators: Pedro Garcia, John Vizhco

- Choose a product or service to be implemented (the "Idea"), which may be a project idea or something different.

We are not only going to provide the product, but also the thought process as we engaged in conversation with the LLM which ended up giving me more ethical implications of the app we had in mind. Sorry for the yap but it is what we wanted to submit since it could be our final project.

The initial motivation for this application stemmed from a frustration I (Pedro) have had from peers, where I would be lending money to friends and wondering whether that money is being used responsibly. When someone owes you money but continues making what appear to be unnecessary purchases, it creates tension and uncertainty about their commitment to repaying the debt. I am a non confrontational type of person so approaches to this problem are not ideal for me as I either have to awkwardly monitor their social media for spending clues, have uncomfortable direct conversations about their purchases, or simply worry in silence while hoping for the best. This creates a genuine need for some form of spending visibility that could provide peace of mind to lenders while helping borrowers demonstrate their financial responsibility.

However, the original concept of monitoring someone's spending because they owe you money presents significant ethical challenges that make it fundamentally problematic as a business proposition. The power dynamic inherent in a debtor and creditor relationship means that "consent" to monitoring may not be truly voluntary and the borrower might feel coerced into agreeing to surveillance in order to maintain the relationship or avoid conflict. This type of financial monitoring could easily cross the line into financial harassment or control, potentially damaging friendships and creating unhealthy relationship dynamics, especially if this was meant for people who are reasonably close to one another. Additionally, determining what constitutes "irresponsible" spending is highly subjective and doesn't account for the borrower's complete financial picture, personal circumstances, or legitimate needs that the lender might not be aware of.

So we thought of a more social aspect to the product where we could use a social accountability model that changes the ethical framework by shifting from surveillance to mutual support. Instead of one person monitoring another due to financial obligation, the revised concept involves voluntary participation among equals who share common financial goals. This could be friends who want to help each other be held accountable for spending habits, or families and roommates who pool money for shared expenses like groceries and housing costs and need a transparent way to ensure their contributions are being used appropriately. This transforms the

application from a tool of oversight into a platform for collaborative improvement, where friends/family help each other develop better spending habits through positive peer pressure and shared accountability. The key differentiator is that everyone in the group is both monitoring and being monitored, creating balanced relationships where the goal is collective financial wellness rather than individual debt recovery.

List similar existing services and why this one is differentiated, or explain your theory about why your Idea doesn't exist yet. It's not enough to say "no one thought of it yet." If it doesn't exist, talk about what convergence of technological or other innovations has occurred to make it feasible now, where it wasn't before.

1. **YNAB (You Need A Budget):** YNAB is a comprehensive budgeting platform that helps individuals and families track expenses, set financial goals, and manage money through detailed categorization and reporting. I have used it to help my spending habits, but it requires a subscription to use each month and The service focuses on proactive budgeting methodology where users manually input transactions and regularly review their spending against predetermined budget categories. However, YNAB is primarily designed for individual or household financial management rather than social accountability, and requires significant manual effort/imputation to maintain accurate records.
2. **Greenlight:** Greenlight provides prepaid debit cards for teens with parental oversight features, allowing parents to set spending limits, approve transactions, and monitor their children's purchases in real time. The platform includes educational parts like savings goals and chore management, designed to teach financial responsibility within a family structure. Unlike social peer accountability, Greenlight operates on a hierarchical parent child control model with the ability to restrict or block spending, and focuses on transaction amounts rather than spending pattern insights.

The primary barrier preventing this social accountability concept has been the nature of financial data APIs, particularly from providers like Plaid where one could create this application. These APIs were designed for established financial services - lending, budgeting, and account aggregation - rather than social features with privacy implications. Plaid's terms of service and pricing structure heavily favor traditional fintech applications, making it prohibitively expensive for startups to access financial data for a more “social” economic model. Additionally, API restrictions around data sharing also make it challenging to create the category only, privacy preserving data flows essential for ethical peer accountability.

In regards to the users/stakeholders, existing financial apps target specific, profitable market segments that don't overlap with this specific use case. Parental control apps like Greenlight serve families willing to pay premium prices (\$15-30/month) for comprehensive oversight, while

individual budgeting tools like YNAB focus on motivated users who invest significant effort in manual financial management.

- **Create a list of stakeholders. Include at least customers, users, regulators, investors, owners, employees, others. It is OK if some of them don't apply, but explain why they don't. Explain their values, motivations and expected behaviors.**

Stakeholder Analysis for Social Spending Accountability App

Customers

Young adults or social circles who pay for the service to access social spending accountability features. Values include financial improvement, relationship transparency, responsible money management, and selective privacy to balance help with spending habits without exposing sensitive financial details like exact amounts or account balances. Expected behaviors include paying subscription fees, inviting friends to join, and engaging with accountability features consistently.

Users

Friend group members and roommates who actively use the app's tracking and sharing features. Values include privacy protection, mutual support, and genuine financial improvement. Motivated by peer support for better spending habits while maintaining personal financial boundaries and autonomy over their data. Expected behaviors include connecting bank accounts, participating in group challenges, providing supportive feedback, and respecting privacy boundaries.

Regulators (AI helped Us with this part)

Financial Regulators (CFPB, FCA): Ensure compliance with financial services laws and consumer protection.

Data Protection Authorities: Enforce GDPR/CCPA compliance for financial data sharing. Values include consumer safety and data security. Motivated by preventing financial abuse and protecting personal information. Expected behaviors include compliance auditing, establishing peer-to-peer financial sharing guidelines, and investigating user complaints.

Investors

Fintech focused venture capital firms maybe. Values include market opportunity, scalable business models, which we discussed earlier. Expected behaviors include due diligence on privacy practices, competitive analysis, monitoring user acquisition metrics, and supporting compliant growth strategies.

Owners

Founding team with equity stakes and board members representing investor interests. Values include ethical product development, sustainable growth, and positive user impact. Motivated by building successful business while maintaining user trust. Expected behaviors include strategic decision-making, ensuring regulatory compliance, and balancing growth with responsible data practices.

Employees

Development Team: Building privacy, preserving social features and secure integrations.

Others

Privacy Advocacy Groups: Monitor financial surveillance and data protection practices. Values include preventing abuse of peer monitoring tools. Motivated by protecting consumers from financial surveillance overreach. Expected behaviors include public scrutiny and advocacy for strong consent mechanisms.

- **Describe one "user journey" or "journey map" per participant on your team. If you are working alone, choose the primary user. If on a team, choose either user and customer (if different) or user and investor (if the same).**

We did a fig jam so that we could create a scenario and compare the values.

Here is the [link](#).

Or -

<https://www.figma.com/board/hqAtLqhQN1qIOwh8bfPVYS/CS1060?node-id=0-1&p=f&t=GYpaECa8pSQGft44-0>

- Submit this part of the assignment as a PDF file called hw2-design.pdf which you should add to your Git repository before submission.

Value Tension:

Between users, Maya and Sarah's journey highlights tensions rooted in different values. Maya prioritizes fairness and efficiency, budget predictability, and transparency without harming friendship, while Sarah emphasizes autonomy in her spending, quality of life, and trust without judgment. The conflict lies in their definitions of fairness: Maya sees it as maximizing value and stretching every dollar, while Sarah views it as equal contribution with freedom of choice. The app resolves these tensions by serving as a neutral ground for transparency and planning, allowing both to protect their values while finding common ground.

Between users and investors, a key tension emerges: Maya and Sarah value privacy protection, relationship preservation, and supportive accountability, only trusting the app if it feels safe and respectful. In contrast, fintech investors prioritize market opportunity, scalable business models, compliance, and growth metrics to justify funding. The balancing act is ensuring the product can scale and deliver investor returns without undermining the privacy-first, relationship-centered values that make it attractive to users in the first place.